ASAHI Caravel Microcatheter

INSTRUCTIONS FOR USE

Caution: Federal (USA) law restricts this device to sale by or on the order of a physician.

Read these instructions carefully before use and observe the Indications for Use, Contraindications, Warnings, Precautions, Malfunctions and Adverse effects and How to Use sections in the Instructions for Use. Failure to do so may result in complications, including serious injury to the patient or death.

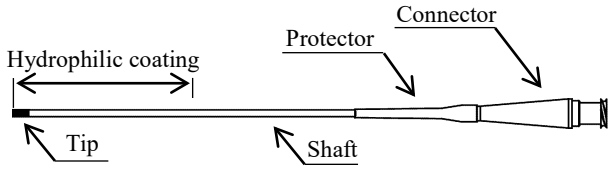
These Instructions for Use apply to the ASAHI Caravel Microcatheter. For specifications of this microcatheter, refer to this product label.

Description

This product is composed of a microcatheter packed in a holder tube. This product is packed in a sterile pack. This sterile pack is packed in a box with this Instructions for Use.

The outer surface of this microcatheter is made of polyamide elastomer coated with hydrophilic polymer to provide high lubricity when the surface is wet. The tip is polyurethane. The inner lumen of the shaft (excluding connector portion) is lined with fluoropolymer layer to facilitate movement of the guide wire and other devices. The tip is radiopaque.

Shape and structure



Indications for Use

This microcatheter is intended to provide support to facilitate the placement of guide wires in the coronary and peripheral vasculatures, and can be used to exchange one guide wire for another.
This microcatheter is also intended to assist in the delivery of contrast media into the coronary and peripheral vasculatures.
Do not use this microcatheter other than for use in the coronary and peripheral vasculatures.

Contraindications

This microcatheter is sterilized with ethylene oxide gas (EOG) and is intended for single use only. Do not resterilize and/or reuse. If reused and/or resterilized, the performance or quality of this microcatheter may be compromised and there is a risk of complications, including infection.

1) Do not modify this microcatheter for any reason. Use of a modified microcatheter may occur damage to the blood vessels and/or accidents.
2) Do not use this microcatheter for patients with following disorders.
   a. Patient with severe heart failure.
   b. Patient with hemorrhagic diathesis or renal failure.
   c. Patient with intractable severe arrhythmia.

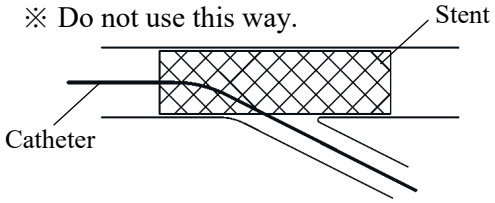
d. Patient with fever or systemic severe infection.
e. Patient with uncompensated heart failure.
f. Patient with severe pulmonary disease.
g. Patient with critical serum electrolyte disorder.
h. Patient with acute myocardial infarction.
i. Patient with blood coagulation disorder or severe change in coagulation ability due to some other causes.
j. Patient who is not eligible for surgical operation or who has exhibited obvious and serious allergic reactions to contrast media or other types of drugs which are necessary for the procedure.
k. Patient who cannot lie on their back on the table for angiography because of congestive heart failure or dyspnea.
l. Patient with mental disease or patient who does not consent to angiography.
m. Patient who is or could be pregnant. (The fetus may be affected by X-rays under fluoroscopy.)
n. Patient with anamnesis of coronary spasm.
o. Patient who cannot be anticoagulated.
p. Any other patient who is judged unsuitable for the procedure by the physician.

3) Do not use this microcatheter in advanced calcified lesion.
4) Do not apply this microcatheter to lesions on the left main trunk, which is not protected by a bypass or collateral circulation.
5) Do not use oleaginous contrast media. (This microcatheter may be damaged.)
6) This microcatheter must not be used for drug infusion. (This microcatheter is not designed for drug infusion and its safety has not been established.)
7) When using a Y-connector, excessive tightening to this microcatheter with the hemostasis valve and manipulation with a tightened Y-connector must be avoided. (This microcatheter may be damaged.)

Warnings

- Do not rotate this microcatheter in any situation. (Rotating this microcatheter may cause damage to the microcatheter, and may also damage the blood vessel.)
- It is recommended that this microcatheter only be used at a medical institution capable of promptly performing an emergency surgical operation. Perform the emergency operation promptly when needed, for example, in the case of an accidental patient injury during a procedure using this microcatheter.
- This microcatheter must be used only by a physician who is fully trained in PTCA and/or PTA.
- If any resistance or something abnormal is felt when operating this microcatheter, do not continue the manipulation while the causes are unclear. If it is suspected that this microcatheter is not operating correctly, avoid excessive manipulations, and carefully remove the entire catheter system while paying full attention to avoid complications. (Continuing the manipulation while the cause of the problem is not identified may cause damage to this microcatheter, and damage the blood vessel.)
- This microcatheter must always be operated under high-resolution fluoroscopic guidance. Particular attention should be paid when inserting or withdrawing this microcatheter into or through stenotic areas, and narrower vessels than the microcatheter. (Abrasion may result in damage of this microcatheter. This may cause vascular injury and perforation.)

- Do not insert or withdraw this microcatheter into or through stent struts.



- Do not insert the guide wire by force or advance it rapidly when this microcatheter is damaged. Such manipulations may cause damage or perforation of the blood vessel.
- Always advance the guide wire ahead of this microcatheter before attempting any manipulation of the microcatheter. (If the guide wire is not advanced ahead of this microcatheter, the microcatheter may be damaged , and the blood vessel may be damaged or perforated.)
- Do not use a power injector with this microcatheter.
- Injection pressure must not exceed 2070 kPa (the maximum injection pressure) when injecting contrast media. (Exceeding the maximum injection pressure may cause damage to this microcatheter.)
- When infusing contrast media, this microcatheter must be operated under high-resolution fluoroscopic guidance, with confirming that the contrast media is being infused from the tip of the microcatheter. If the contrast media is not being infused, infusion must be stopped and this microcatheter must be replaced with new one.（If this microcatheter lumen is blocked, the microcatheter may be damaged even at not more than the maximum injection pressure.）
- Agents containing organic solvent such as alcohol must not be used either alone or concurrently. These agents must not be used for immersing or wiping this microcatheter. (This microcatheter may be damaged or lose its lubricity.)
- The patient may suffer from subacute thrombosis, vascular complications, or bleeding complications by using this microcatheter. Therefore, it should be well examined if the intervention procedure will be applicable for the patient.
- Repeated insertion and withdrawal of this microcatheter may lead to deterioration of the hydrophilic coating. (Continuous use of this microcatheter with deteriorated hydrophilic coating may cause vascular damage. This may also increase the risk of trapped tip, and cause a damage of the tip.)
- Comply with instructions, precautions, and warnings described in the Instructions for Use supplied with medical devices (Sheath introducer kit, Angiographic catheter, Guide catheter, Guide wire, etc.) used together with this microcatheter.
- Do not manipulate the stopcock of the guide catheter when this microcatheter is inserted in the guide catheter fitted with a stopcock. It may cause damage of this microcatheter or the guide wire.

Precautions

1. Contraindication and prohibition

Do not use if the package is opened or damaged. Always open the package just prior to use. Use aseptic technique in handling and using this microcatheter.

2. Important precautions

1) Prior to use, check all devices, including this microcatheter, and confirm that they function normally. Check also if the product is not damaged during transportation. Do not use if the package and/or the product is suspected to be damaged.

2) Prior to use, confirm that this microcatheter is compatible with the interventional device to be used.
3) Prior to use, make sure that specifications of this microcatheter are suitable for the purpose of use and the procedure.
4) Use by the expiration date indicated on the label of the product package.
5) When inserting the guide wire into this microcatheter which is already placed in the blood vessel, carefully operate the guide wire not to damage the microcatheter at the bend sections.
6) Confirm that this microcatheter does not have any damage before injecting contrast media.
7) Do not use this microcatheter for the purpose other than described in the Indications for Use written in this Instructions for Use.
8) Select the appropriate size of guide catheter and guide wire to use in combination with this microcatheter. (See the label of product package)
9) Operate this microcatheter carefully to avoid damage, especially when inserting the microcatheter into the guide catheter.
10) Check the patient's condition before the procedure. Provide appropriate anticoagulant therapy.
11) When infusing contrast media, read carefully the Instructions for Use provided with such contrast media and comply with instructions, precautions, and warnings.
12) The surface of this microcatheter is coated with hydrophilic polymer. Flush the surface and the lumen of this microcatheter continuously with heparinized and sterilized saline during its use to maintain lubricity.
13) When inserting or exchanging this microcatheter, flush the lumen of the guide catheter and the microcatheter system continuously with heparinized and sterilized saline.
14) Flush the lumen of this microcatheter sufficiently with heparinized and sterilized saline especially after injecting contrast media.
15) Discontinue injection if irregular resistance is felt at the syringe. This microcatheter may be damaged or blocked. Excessive pressure may cause damage of this microcatheter.

3. Other precautions

Take preventive measures against infection after use. Discard this microcatheter as medical waste.

Malfunctions and Adverse effects

When using this microcatheter, the following malfunctions and adverse effects may occur. A serious adverse effect may lead to severe complication or death. However, such malfunctions and adverse effects are not limited to these.

1) Malfunctions
   - Damage
     - ・Kink
     - ・Bend
     - ・Separation
     - ・Burst
     - ・Damage to the hydrophilic coating
   - Withdrawal difficulty
   - Insertion difficulty
   - Trap with guide wire

2) Adverse effects
   - Death
   - Infection
   - Vessel dissection or perforation

- Hemorrhage or hematoma
- Air embolism
- Embolization or fragmentation of thrombotic or atherosclerotic material
- Infarction
- Embolization or vascular damage caused by residue
- Ischemia
- Arrhythmias, including ventricular fibrillation
- Spasm of blood vessels
- Acute vessel closure
- Aneurysma (false / dissecting)
- Hypotension
- Allergic reaction

How to Use

This microcatheter shall be selectively inserted in blood vessel over a guide wire.

<A> Instructions for use as an infusion catheter

1) Take out the holder tube containing this microcatheter from the sterile pack.
2) Inject the heparinized and sterilized saline into the holder tube through the flush connector by using a syringe. Ensure the ejection of heparinized and sterilized saline from distal end of the holder tube to make sure the holder tube is filled with heparinized and sterilized saline.
3) Remove this microcatheter from the holder tube, and check the surface of the microcatheter for sufficient lubricity. If any resistance is felt when withdrawing this microcatheter from the holder tube, inject additional heparinized and sterilized saline into the holder tube to lubricate the microcatheter.
4) Flush the lumen of this microcatheter removed from the holder tube with the heparinized and sterilized saline by using a syringe. Fill the lumen of this microcatheter with the heparinized and sterilized saline.
5) Insert the appropriate guide wire (indicated on the label of product package) into this microcatheter and advance carefully.
6) Insert the guide catheter into the patient's blood vessel according to standard catheter procedure.
7) Insert this microcatheter and the guide wire as a unit into the guide catheter, from its hemostatic adaptor (Y-connector etc.), which is inserted in the patient's vessel.
8) Advance this microcatheter and the guide wire under fluoroscopy until it reaches 2 to 3 cm proximal of the tip of the parent guide catheter.
9) Advance only the guide wire to the distal of the blood vessel.
10) After loosening the hemostasis valve of the Y-connector, if this microcatheter is hindered by a stenosed area, and/or when enough guide wire support is not obtained, secure tightly both the guide wire and the guide catheter. Then, advance this microcatheter slowly along the guide wire and observe the movement of the tip of the microcatheter to determine if the tip passes through the stenosed area.
11) Before injecting contrast medium, withdraw the guide wire. Connect a syringe to the connector, and manually inject contrast medium .
12) When thrombus adhesion is expected during the procedure, connect the hemostatic adaptor to the connector of this microcatheter, and inject heparinized and sterilized saline from the port of the hemostatic adaptor by using a syringe, or connect a pressured bag with heparinized and sterilized saline for continuous drip to prevent thrombus from adhering to the microcatheter.

13) Attach an extension wire to the proximal end of the guide wire or use a 3M or longer guide wire before removing this microcatheter.
14) When removing this microcatheter, loosen the hemostatic valve of the hemostatic adaptor. Remove this microcatheter while keeping the guide wire stable in the blood vessel. (When this microcatheter is removed, check the position of the guide wire under fluoroscopy. Also, if any resistance is felt during the removal of this microcatheter, remove all devices including the parent microcatheter and the guide wire.)
15) After removal of this microcatheter, tighten the hemostatic valve of the hemostatic adaptor.

<B> Instructions for use as a support catheter

1) Take out the holder tube containing this microcatheter from the sterile pack.
2) Inject the heparinized and sterilized saline into the holder tube through the flush connector by using a syringe. Ensure the ejection of heparinized and sterilized saline from distal end of the holder tube to make sure the holder tube is filled with heparinized and sterilized saline.
3) Remove this microcatheter from the holder tube, and check the surface of the microcatheter for sufficient lubricity. If any resistance is felt when withdrawing this microcatheter from the holder tube, inject additional heparinized and sterilized saline into the holder tube to lubricate the microcatheter.
4) Flush the lumen of this microcatheter removed from the holder tube with the heparinized and sterilized saline by using a syringe. Fill the lumen of this microcatheter with the heparinized and sterilized saline.
5) Insert the appropriate guide wire (indicated on the label of product package) into this microcatheter and advance carefully.
6) Insert the guide catheter into the patient's blood vessel according to standard catheter procedure.
7) Insert this microcatheter and the guide wire as a unit into the guide catheter, from its hemostatic adaptor (Y-connector etc.), which is inserted in the patient's vessel.
8) Advance this microcatheter and the guide wire under fluoroscopy until it reaches 2 to 3 cm proximal of the tip of the parent guide catheter.
9) Keeping the guide wire ahead of this microcatheter, advance the microcatheter under fluoroscopy until it is close to the stenotic area. Advance the guide wire carefully until it passes the target area. Continue advancing the guide wire as distal as possible into the blood vessel, and once it is placed there, check the position by angiography from the guide catheter. The position of the guide wire must be checked by angiography from multiple angles to confirm that the guide wire is definitely inserted into the target blood vessel.
10) After loosening the hemostatic adaptor, hold the guide wire and guide catheter firmly. Then advance this microcatheter gradually along the guide wire until the distal end has passed through the stenotic area, using the tip of the microcatheter as a guide. (Procedures inside the blood vessel should be conducted with care, because this microcatheter is hydrophilic coated.)
11) Attach an extension wire to the proximal end of the guide wire or use a 3M or longer guide wire before removing this microcatheter.
12) When removing this microcatheter, loosen the hemostatic valve of the hemostatic adaptor. Remove this microcatheter while keeping the guide wire stable in the blood vessel. (When this microcatheter is removed, check the position of the guide wire under fluoroscopy. Also, if any resistance is felt during the removal of this microcatheter, remove all devices including the parent microcatheter and the guide wire.)
13) After removal of this microcatheter, tighten the hemostatic valve of the hemostatic adaptor.

Storage Condition

Do not keep this product in a bent and/or heavily-loaded condition. This product must be kept out of water. Store in a cool, dark, and dry place.

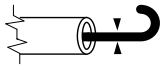
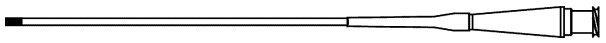
Expiration date

The expiration date is indicated on the label of product package.

Contents

1 set per package

Symbols

	Recommended guide wire diameter		Maximum injection pressure
	Inner diameter		Minimum guide catheter I.D.
		Catheter	

Liability Disclaimer

By no means shall "ASAHI INTECC CO., LTD. and its affiliated companies" (hereinafter referred to as the "Company") be liable for accidents, personal injuries, adverse effects due to any improper use of the product(s) or any other use inconsistent with these instructions. In no event shall the Company be liable for any damages either (i) arising out of storage of the product(s) after the shipment from the Company or (ii) due to selection of patients, surgery techniques, or any other medical activities by the medical institution that uses the product(s).

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